

FINNER, Ireland

WMO: 039780

Lat: 54.4939N

Lon: 8.2431W

Elev: 35

StdP: 100.91

Time Zone: 1.00 (IRL)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.1	-0.5	-4.6	2.6	-0.1	-3.1	2.9	1.5	17.7	6.1	15.9	7.2	2.4	110	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	21.9	17.2	20.1	16.3	18.8	15.6	18.0	20.6	17.0	19.2	16.2	18.1	4.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.1	19.3	16.1	11.5	18.2	15.3	10.9	17.3	50.9	21.0	48.0	19.3	45.5	18.3	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.6	11.1	DB	-3.5	25.0	2.2	2.0	-5.1	26.4	-6.4	27.6	-7.6	28.7	-9.2	30.1	(4)
(5)			WB	-4.2	20.2	2.0	1.2	-5.6	21.1	-6.8	21.8	-7.9	22.5	-9.4	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	6.0	5.8	6.9	8.7	11.2	13.7	14.9	14.8	13.4	10.8	7.9	6.2		
	DBStd	4.20	2.81	2.67	2.67	2.42	2.56	2.29	1.96	1.48	2.02	2.50	2.71	3.20		
	HDD10.0	628	126	119	102	52	14	1	0	0	1	21	71	121		
	HDD18.3	3031	382	351	356	289	223	141	108	111	149	234	312	376		
	CDD10.0	644	2	2	5	13	51	112	153	147	103	45	9	3		
	CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH23.3	26	0	0	0	0	3	11	12	1	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	1	1	0	0	0	0	0		
(14) Wind	WSAvg	5.3	6.2	5.7	5.7	5.1	4.9	4.6	4.5	4.8	5.1	5.1	5.6	5.8		
(15) Precipitation	PrecAvg	1398	150	116	101	91	89	86	104	108	104	140	151	159		
	PrecMax	1634	215	250	194	155	187	174	174	212	168	267	298	305		
	PrecMin	1049	19	38	40	35	20	28	52	18	17	24	82	51		
	PrecStd	166	52	57	42	33	42	36	35	43	39	64	48	68		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.3	15.7	18.5	23.3	24.5	24.8	21.6	20.7	17.8	15.2	14.0		
		MCWB	11.5	10.9	11.1	12.4	17.0	17.5	19.2	17.8	16.8	15.2	13.4	12.2		
	2%	DB	11.8	12.0	13.3	15.7	19.9	21.4	21.6	19.6	19.0	16.2	13.7	12.8		
		MCWB	10.6	10.2	10.3	11.4	15.2	16.1	17.8	16.6	15.9	14.1	12.3	11.1		
	5%	DB	11.0	11.0	12.0	14.0	17.4	19.4	19.7	18.4	17.7	15.2	12.6	11.9		
		MCWB	9.7	9.5	9.7	10.6	13.4	15.4	16.7	15.9	15.4	13.4	11.4	10.5		
	10%	DB	10.1	9.9	10.9	12.6	15.5	17.6	18.2	17.5	16.6	14.3	11.7	10.8		
		MCWB	8.9	8.5	8.9	9.8	12.3	14.5	15.6	15.2	14.6	12.6	10.4	9.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.5	11.9	13.4	17.6	18.6	20.2	18.6	17.8	15.7	13.8	12.4		
		MCDB	12.8	12.6	14.4	16.6	21.9	23.2	23.9	20.5	19.1	17.2	14.9	13.6		
	2%	WB	10.8	10.5	10.8	12.1	15.6	17.0	18.3	17.3	16.6	14.6	12.5	11.4		
		MCDB	11.7	11.7	12.6	14.8	19.4	20.5	20.9	18.8	18.1	15.9	13.4	12.5		
	5%	WB	9.9	9.6	10.0	11.0	14.1	15.9	17.1	16.5	15.7	13.7	11.6	10.6		
		MCDB	10.8	10.8	11.7	13.5	16.6	18.7	19.2	17.9	17.2	15.0	12.5	11.7		
	10%	WB	9.1	8.6	9.1	10.1	12.7	14.9	16.0	15.7	14.9	12.9	10.7	9.6		
		MCDB	10.0	9.8	10.6	12.2	14.9	17.2	17.8	17.0	16.4	14.1	11.7	10.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.9	5.8	6.4	6.6	6.1	5.5	5.1	5.4	5.2	4.6	4.5		
		MCDBR	4.9	4.9	7.0	8.6	9.9	9.5	7.9	6.4	6.7	5.5	4.8	5.3		
		MCWBR	4.9	4.4	4.9	5.2	5.8	5.6	5.0	4.2	4.7	4.2	4.6	4.9		
	5% WB	MCDBR	4.7	4.9	6.2	7.4	8.8	8.4	7.4	5.6	5.9	5.0	4.6	5.0		
		MCWBR	4.9	4.6	4.9	5.0	5.7	5.3	5.0	4.4	4.7	4.4	4.7	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.347	0.353	0.374	0.383	0.391	0.377	0.380	0.364	0.362	0.349	0.338	0.336		
	taud		2.197	2.277	2.271	2.290	2.311	2.388	2.389	2.446	2.443	2.398	2.263	2.170		
	Ebn at Noon		562	718	792	841	854	870	860	855	805	721	575	483		
	Edh at Noon		65	83	103	115	119	112	110	98	87	72	59	54		
(44) All-Sky Solar Radiation	RadAvg		0.56	1.25	2.33	3.78	4.83	4.93	4.41	3.69	2.61	1.50	0.70	0.41		
	RadStd		0.06	0.12	0.20	0.31	0.41	0.44	0.30	0.30	0.16	0.12	0.06	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page